TrailBuilder

Expert Review Dossier

For: a mechanic / suspension tech / bike-industry engineer reviewing the compatibility engine.

Prepared: 2026-07-07 · engine state: 19 rule areas + sub-checks, 198 tests, 330 catalog parts (74 verified).

Time to review: ~2–3 hours with this document; no code reading required.

Why you're being asked

TrailBuilder tells people whether bike parts fit together. Its tests prove the engine is

self-consistent — they cannot prove the rules are *right about the real world*. That is what you're for. Every rule below was built from manufacturer documentation, but manufacturer docs are incomplete, shop practice diverges from them, and we are not mechanics.

The bar the tool lives by: a wrong verdict is worse than a missing rule — in both directions. A false "won't fit" steers someone away from a working build; a false "fits" sells someone a part that doesn't mount. When in doubt the engine stays silent, and silence is disclosed ("No conflicts found", never "All compatible").

What we want from you, per rule

1. **Is the claim true** as stated, in your shop experience — not just per the docs?
2. **Is the tier right?** Error = won't fit, period. Warning = fits/works but check something first. Info = FYI. (Warnings show as a yellow dot before a part is added.)
3. **Is the direction right?** Several rules fire only one way on purpose — those are the likeliest places we've encoded a subtle wrong belief.
4. **What common misfit did we not check at all?** The gaps list at the end is what we know we're missing; the dangerous gaps are the ones we don't.

Mark each rule [Y] (correct), [N] (wrong — say why), or ~ (right idea, wrong tier/wording/scope).

Rule-by-rule

1. Wheel size configuration

Claims: front parts (fork/wheel/tire) must agree on size; rear parts likewise; the front/rear combo must be one the frame supports: 29/29, 27.5/27.5, or mullet (29 F / 27.5 R). With no frame picked, a pair matching no configuration (e.g. 27.5 F / 29 R) is rejected.

Tier: error. **Basis:** frame makers' published wheel-size options per model.

Ask: is hard-rejecting reverse-mullet correct, or does some niche legitimately run it?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

2. Axles

Claims: fork axle must match front hub (everything current is Boost 15×110); frame rear spacing must match rear hub (Boost 148 vs SuperBoost 157 mismatch = error).

Tier: error. **Note:** the front check can't currently fire (every catalog part is Boost 110) — it's pinned by synthetic tests for the day the vocabulary widens.

Ask: SuperBoost frame + Boost rear wheel = error with no adapter path — correct?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

3. Drivetrain family — three checks

- **3a. One system, one speed count** across shifter/derailleur/cassette/chain (SRAM Eagle / SRAM Transmission / Shimano 12 are mutually incompatible). Error.
- **3b. Actuation:** a cable trigger cannot drive an AXS (wireless) derailleur and vice versa, even within the same "system" — Eagle cassettes/chains are genuinely shared between mechanical and AXS, so the check is a separate field, not a system split. Error.
- **3c. Chainring standard, one-directional:** a Transmission Flattop chain on a non-T-Type chainring is an error (SRAM documents unique link shape/pin/rollers — support.sram.com, corroborated by Wolf Tooth and e*thirteen). The REVERSE stays silent on purpose: SRAM documents T-Type rings as backward-compatible with Eagle chains. Cranks sold without rings (eeWings, Race Face armsets) get an info, not an error. **Ask:** is 3c's one-directionality right? Any real-world T-Type ring + Eagle chain problems?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

4. SRAM Transmission needs UDH

Claims: a direct-mount (UDH) Transmission derailleur on a non-UDH frame = error; with no frame picked it's a heads-up info, not a red.

Tier: error / info. **Basis:** SRAM's UDH requirement; per-frame UDH status web-sourced for all 21 frames (e.g. the stock RAAW Madonna V2.2 is NOT UDH — RAAW sells a retrofit kit, so the bare frame correctly rejects it).

Ask: any edge cases (aftermarket hangers, frames converted)?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

5. Cassette range vs derailleur capacity — deliberately one-sided

Claims: cassette maxCog > derailleur max = error (10-52 cassette on a 50T-max mech). A 52T-capable derailleur with a 10-50 cassette is **silent on purpose** — an auditor proposed erroring it and verification found SRAM documents the opposite (520% derailleurs are backward-compatible with 10-50). The refutation is recorded so it never gets "fixed" in.

Ask: confirm the one-sidedness; any b-screw/setup caveat worth a warning?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

6. Cassette freehub vs rear wheel

Claims: exact-match required (XD / MicroSpline / HG). Known modeling limitation: freehub bodies are actually SWAPPABLE on most platforms, but the engine treats them as fixed per wheel row; we mitigate by cataloging the maker's offered configs as separate rows (e.g. the DT E 1900 27.5 exists in both MicroSpline and XD rows).

Tier: error. **Ask:** is "needs the XD version of this wheel" (separate row) honest enough, or should the error message mention body swaps explicitly?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

7. Bottom bracket — advisory only

Claims: no BB category exists yet; every crank×frame pair gets an info naming the spindle standard (DUB / 24 mm / 30 mm / e*13 P3) and the frame shell (BSA73 / PF92 / T47) and saying "BB sold separately". Nothing is ever blocked.

Ask: every current spindle×shell pair is servable with a real BB — will that stay true? Which pairs deserve a real warning (e.g. 30 mm spindle in PF92)?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

8. Brake caliper mounts

Claims: caliper mount must match fork/frame mount. Currently every part is post-mount, so the check can't fire (pinned synthetically). **Post-mount SIZE (PM160/180/200 native) is NOT modeled** — the protection comes from rule 10's min/max rotor checks instead.

Ask: is delegating PM-size protection to rotor min/max sufficient, or does it leak?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

9. Rotor interface vs hub — direction-aware

Claims: a Center Lock rotor on a 6-bolt hub = error, no adapter exists. A 6-bolt rotor on a Center Lock hub = **warning** naming the adapter (Shimano SM-RTAD05-class, one-piece rotors only) — an everyday shop fix, so a hard red was a false "won't fit".

Ask: the adapter caveat says one-piece rotors only (not Hope floating / two-piece) — right? Is warning the right tier or should adapters be a quieter info?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

10. Rotor size — max (warning) and native-mount minimum (error)

Claims: rotor above the frame/fork's published max = warning ("exceeds the max"). Rotor BELOW a fork's native post-mount size = **error**: adapters only space calipers UP, so a 180 rotor on a 200-native ZEB/Domain mount physically cannot work (sram.com states "Minimum Rotor Size: 200mm" for those forks; populated only where fetched).

Ask: should over-max be promoted to error? It's a manufacturer structural limit, but it *mounts* — we kept it a warning and made warnings visible as yellow dots. Judgment call.

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

11. Steerer / headset

Claims: fork steerer must match frame (everything current is tapered 1.5–1.125; check pinned synthetically). **Ask:** anything worth modeling before 1.8" steerers arrive?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

12. Fork travel vs frame — rated max (warning) + approved minimum (dormant)

Claims: fork travel above the frame's rated max = warning ("exceeds the frame's rated max" — worded as rated, not recommended, because it's warranty-relevant). UNDER-forking warns only when a frame carries a maker-published minimum (none does yet — deliberately no heuristic: high-pivot frames like the Dreadnought ship 154 mm travel with 170 forks, so a travel-based guess would false-fire).

Ask: ~1° head-angle steepening per 20 mm is the number we quote — fair? Tier right?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

13. Dropper vs seat tube — direction-aware + insertion nudge

Claims: post bigger than the tube = error (physically impossible). Post SMALLER than the tube = warning naming the exact reducing shim ("31.6-to-30.9 mm, sold separately") — shop-approved everyday practice (Wolf Tooth/Problem Solvers; PNW's own sizing guide endorses shimming). Posts with ≥200 mm drop additionally get an INFO pointing at the maker's insertion calculator, because the app has no frame-size concept yet and insertion depth is the most common real dropper misfit we cannot check.

Ask: shim = warning, too-big = error — right? Is the 200 mm info threshold sensible?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

14. Tire vs rim width + fork crown clearance

Claims: tire wider than the wheel's max = warning (active — 2.6" tires exist in the catalog against 2.5"-max rims). Tire wider than the FORK's published chassis clearance = warning, dormant until forks carry sourced data (Fox/RockShox publish per chassis).

Not modeled: too-NARROW tire on a wide rim (thresholds are fuzzy/standards-dependent — deferred pending sourcing).

Ask: rim maxTire values are partly sample guidance, not maker data (flagged per row) — how much do you trust ETRTO-style width guidance as a default?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

15. Handlebar / stem clamp

Claims: 31.8 vs 35 mm mismatch = error. No known caveats; both live in the catalog.

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

16. Rear shock fit — direction-aware stroke + coil approval

Claims: eye-to-eye mismatch = error. Mount (std vs trunnion) mismatch = error. Stroke LONGER than the frame's spec = error (over-rotation / bottom-out contact). Stroke SHORTER with matching eye+mount = **warning** quantifying reduced travel ("~158 mm instead of 165 mm") — makers sell the same body in multiple strokes and RockShox supports stroke reduction, so a red was a false "won't fit". A coil shock on a frame whose maker states "not coil-compatible" warns (dormant — populated only from maker statements, never leverage-curve guesses). Hardtails cleanly reject any shock.

Ask: travel is estimated linearly (frame travel × stroke ratio) — close enough for a warning? Should shorter-stroke mention stroke spacers specifically?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

17. Frame + shock bundling / OEM-only

Claims: package-only frames (e.g. Specialized Enduro) warn when you pick a different shock ("you'll still pay for the bundle"); OEM-only shocks error on the wrong frame and give an info when no frame is picked yet.

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

18. Rear tire vs frame clearance

Claims: rear tire wider than the frame's published max = warning. ACTIVE on 10 catalog frames (8 models) with manufacturer-published clearances (Madonnas 2.6, Slash 2.5, Megatower 2.5 stock, Spire 2.6, SB160 2.6, HD6 2.5, Reign 2.5, Dreadnought 2.6); frames without a published max stay silent by design.

Ask: Megatower's 2.5 is the "stock setting" figure (flip-chip dependent) — is per-setting clearance worth modeling, or is stock-setting honest enough?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

19. Shifter clamp vs brake-lever integration

Claims: a lever-integrated shifter (I-Spec EV/II/B, MatchMaker — ships with no bar clamp of its own) paired with brakes that accept none of its standards = warning ("needs the band-clamp version or a ShiftMount-style adapter"). I-Spec generations are treated as mutually incompatible; a lever may accept several standards; one matching lever is enough.

Ask: generation matrix right? Adapter-tier wording fair?

Mechanic verdict: [ ] correct [ ] wrong (say why below) [ ] right idea, wrong tier/wording

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Deliberate NON-rules — please confirm, don't "fix"

1. **SuperBoost frame + Boost-chainline crank: NO error.** Commonly ridden (e.g. Firebird with Boost cranks); a naive chainline rule would be a false red. Pinned by a test so a future session can't add it. The `chainline` field is display-only bait. **Agree?**
2. **52T-max derailleur + 10-50 cassette: NO error** (see rule 5 — SRAM documents backward compatibility; an auditor's proposed rule was refuted and recorded).
3. **Ride discipline (XC/trail/enduro/DH) NEVER gates fit** — a DH tire physically fits an XC bike. Disciplines are filter/annotation only.
4. **Pedals: no compat rules** — 9/16" thread fits every crank.

What "verified" means in the data

- **Verified (74 of 330):** every spec set from a **fetched manufacturer page/document** (never search summaries — they've been wrong about shock sizes, seatpost diameters and rotor maxes in this exact catalog), plus a dated source URL. Weights for makers who publish none (Shimano, SRAM rotors, Fox shocks) may come from a reputable third-party **measured** figure for the exact SKU — flagged `measured` with its own URL; retailer listings are rejected. Everything else is clearly badged **sample data** in the app.

- Verification keeps catching fabricated sample specs — four tire sizes that don't exist, weights off by 50–200 g, a fictional "Kryptotal" that's really two treads. Assume unverified rows contain more of the same.

Known gaps — please rank by real-world danger

Gap	Status
Dropper insertion depth vs frame size	Info nudge only; real check needs a frame-size concept
Bottom bracket as a real part/rule	Advisory info only (see rule 7)
Tire too narrow for rim	Not checked (thresholds unsourced)
Under-forking	Rule exists, dormant — no frame publishes a minimum yet
Coil approval per frame	Rule exists, dormant — needs maker statements
Front tire vs fork crown	Rule exists, dormant — needs fork clearance data
Post-mount native size	Not modeled (delegated to rotor min/max)
Chain length / ring size vs derailleur capacity	Not checked
Seatpost insertion vs curved/interrupted seat tubes	Not checked
E-bike-specific ratings	Not modeled at all

The most valuable thing you can tell us is the misfit you see weekly that isn't anywhere in this document.

How to report

Pre-deploy: annotate this file, or use the app's ■ **Report a wrong verdict** button (any build state can be flagged; it produces a copyable reproduction report with a share link). Post-deploy those reports become pre-filled GitHub issues.